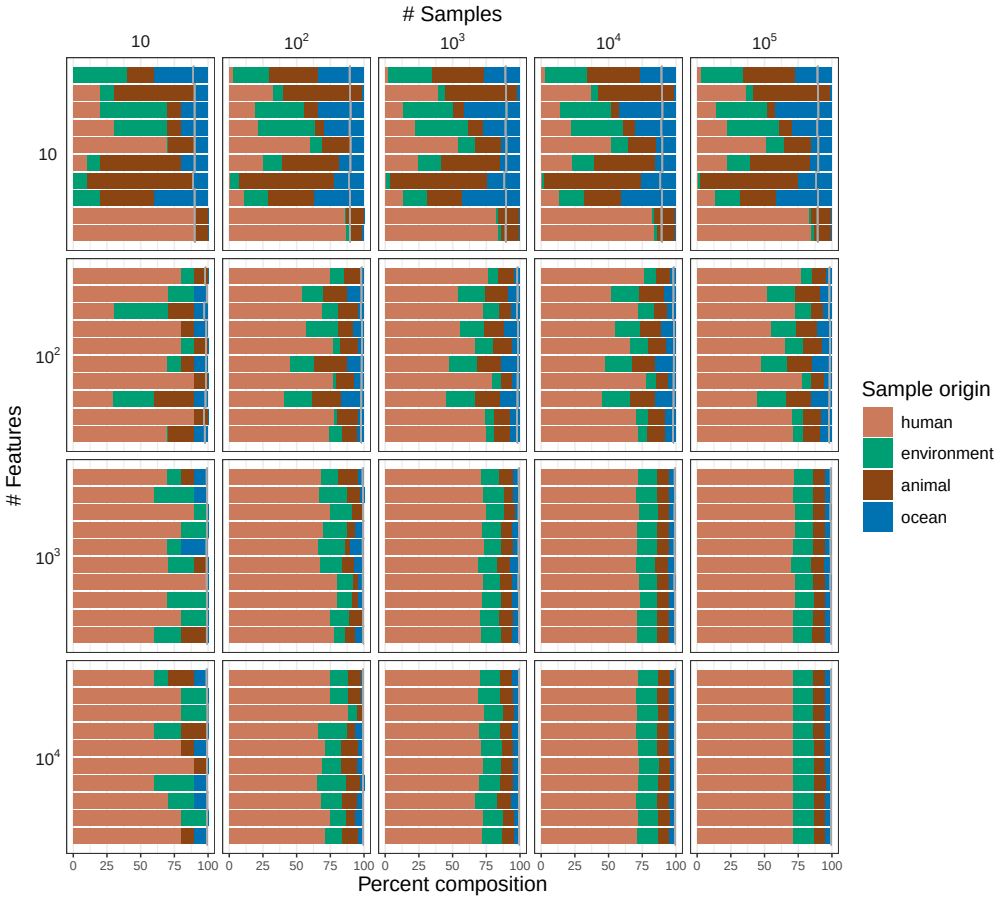
Supplementary Information



Supplementary Figure 1: **Sample composition of benchmarking subsets.** The barplot depicts the percent composition by sample origin for the subsets of the Metalog dataset used in the benchmarking analysis[1]. Each facet corresponds to one combination of feature (rows) and sample sizes (columns). The bars within each facet represent the ten different replicates generated using random states 0-9. Sample origin is encoded by bar color, and the percent sparsity of each subset is marked with gray vertical lines.

Supplementary references

[1] Kuhn, M. *et al.* Metalog: curated and harmonised contextual data for global metagenomics samples. *Nucleic Acids Research* **54**, D826–D834 (2026).